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# GROUP 2- CO2 FOOTPRINT TRACKER

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Guidance
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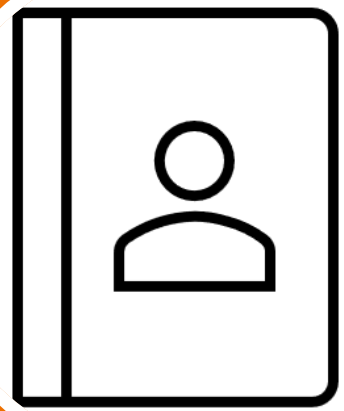
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# Agenda

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01 | Introduction

02 | Project Definition

03 | Key Features

04 | System Architecture

05 | UML Diagrams

06 | Demo

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# Introduction

- **Motivation:** Increased CO2 level resulting in global warming (average temperature rise by 1.17 C in last 70 years)**.**
- **Purpose**: Empower individuals to measure and monitor their carbon footprints.
- Configure lifestyle attributes (e.x. household, mobility).
- Record daily activities and their $CO_2$ footprints.
- Analyse footprints by category and time period.
- Promote sustainable living and environmental awareness.
- Educate users on the impact of their lifestyle choices.

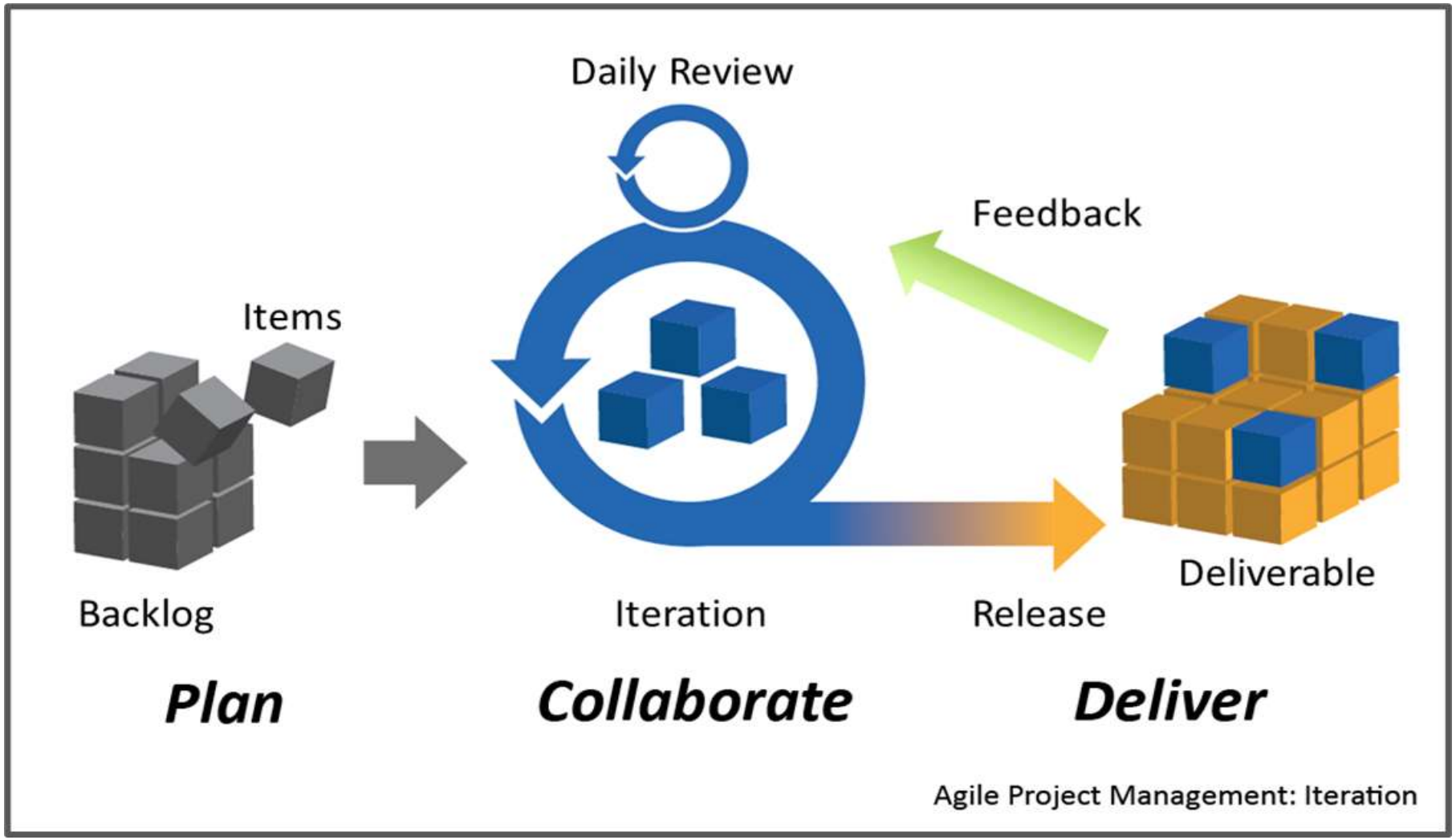
# Project Definition

- The $CO_2$ Footprint Tracker is a web-based application designed to help users calculate, monitor, and track their carbon footprints. By leveraging user-provided data, the application provides personalized insights into $CO_2$ contributions from daily activities such as transportation, energy use, and consumption patterns.
- The project aims to promote environmental awareness and sustainable behavior through an interactive platform offering educational resources, actionable recommendations, and progress tracking.
- Target user would be Environmentally Conscious People, Students & Educators, Fitness Enthusiasts, Travel Enthusiasts, Organizations etc.

# Key Features

- Personalized Configuration
- Activity Logging
- Analysis Dashboard
- Category Insights
- Progress Tracking
- Unique Selling Point: Personalized CO2 tracking with lifestyle-based visualisation.

# System Architecture



# User stories

| |
|---|
| Sprint 1 - Basic Setup, Research and Analysis |
| Sprint 2 - Daily $CO_2$ footprints Input |
| Sprint 3 - Reports and Analytics with Data Visualization |
| Sprint 4 – Testing, Documentation, Knowledge Transfers,  Advanced Features |

# Tools And Technology

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| FRONT-END | |
|---|---|
| JSP | HTML |
| CSS | JAVA SCRIPT |

| BACK-END | |
|---|---|
| spring Boot | JAVA |
| MySQL Server | |

# Springboot RestfulAPI

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- Facilitate seamless communication between frontend and backend.
- Scalable, secure, and standardized data exchange.

# MVC Architecture

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Model-View-Controller is a software design pattern for separating concerns in application development.

- Model :

  - Represents data and business logic.

  - Interacts with the database.

- View :

  - Displays data to the user.

  - Interacts with the controller for updates.

- Controller :

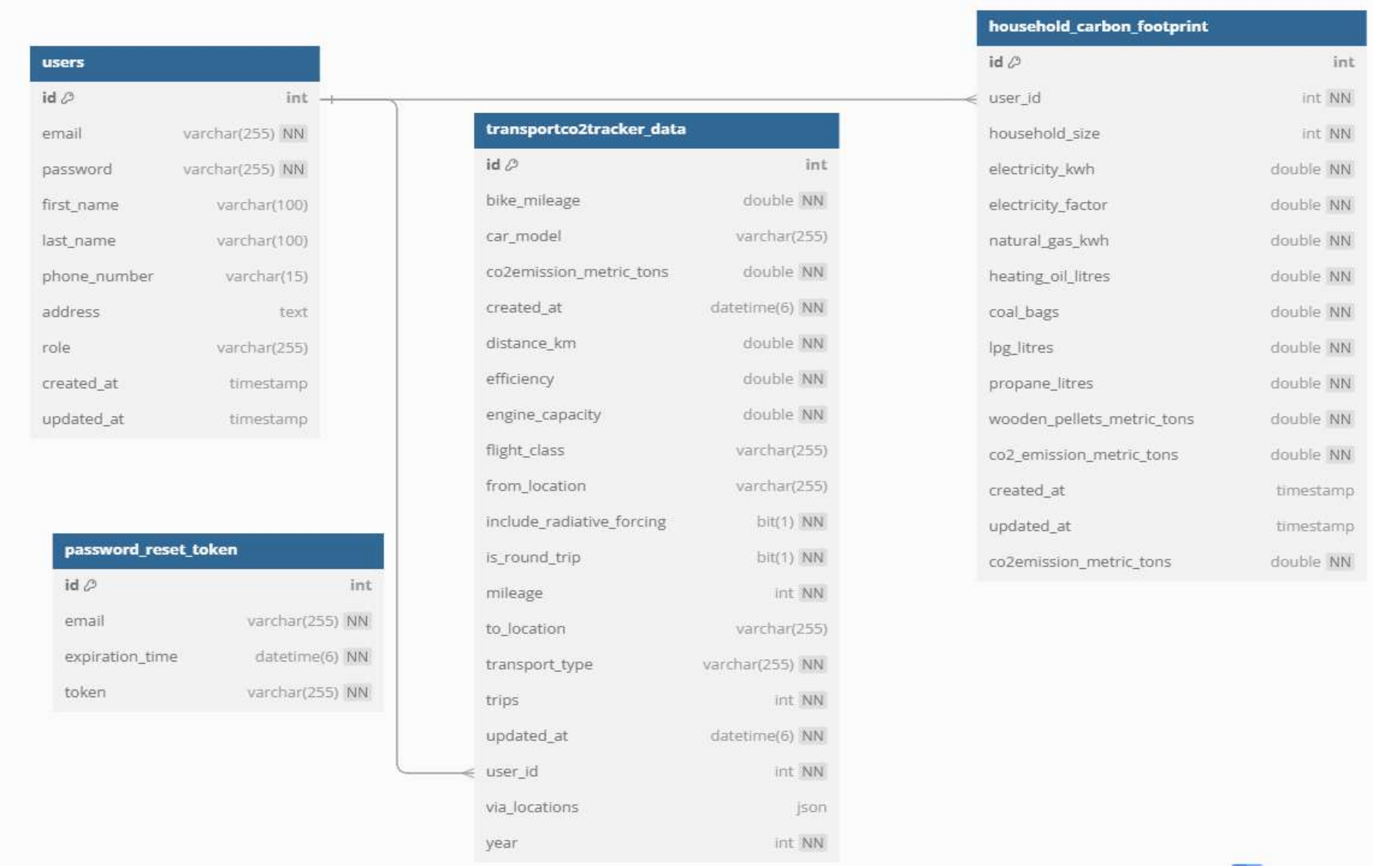
  - Handles user input and orchestrates communication between Model and View.

# UML Diagrams

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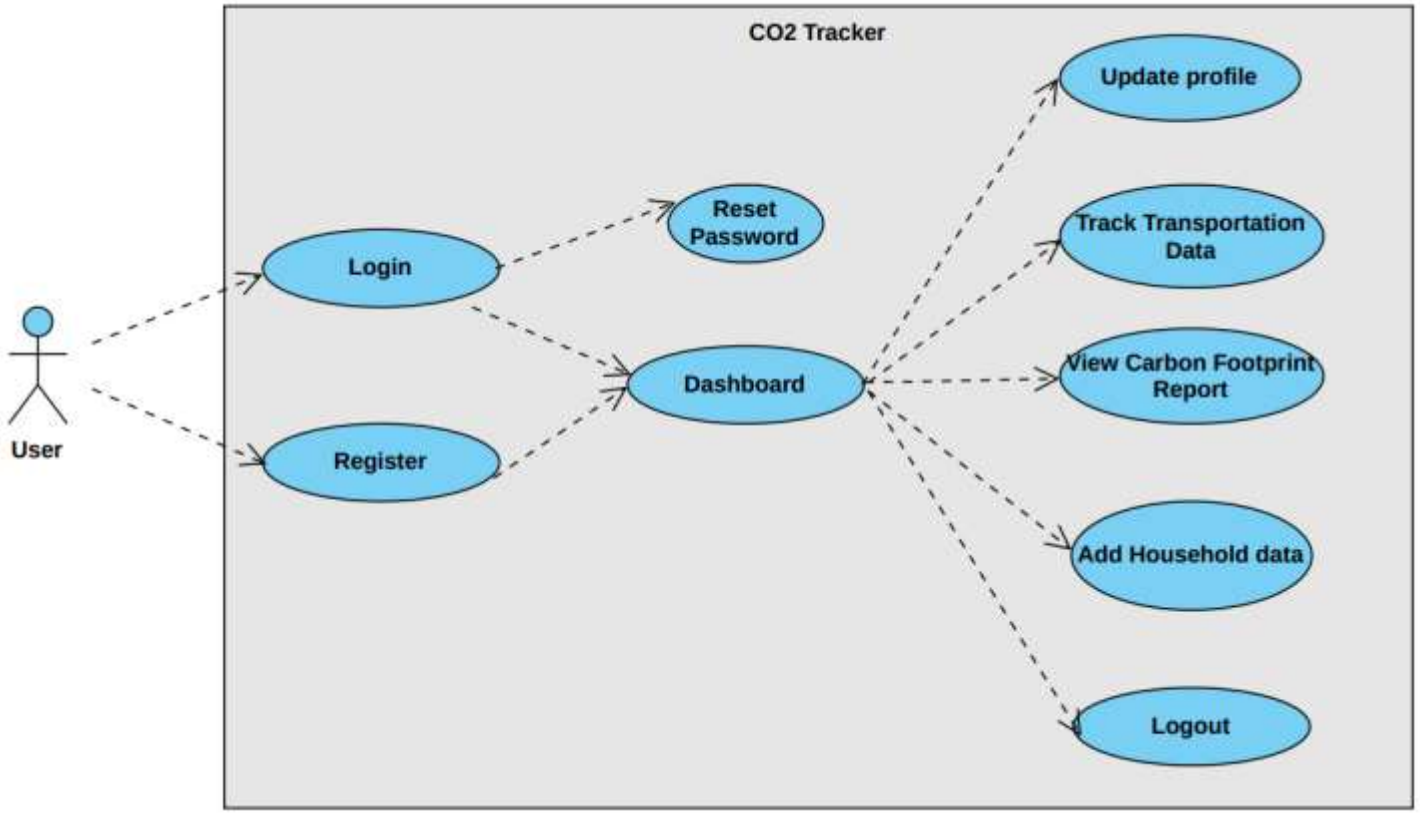
- ER-Diagram
- Use Case Diagram
- Activity diagram

# Diagrams
## ER Diagram



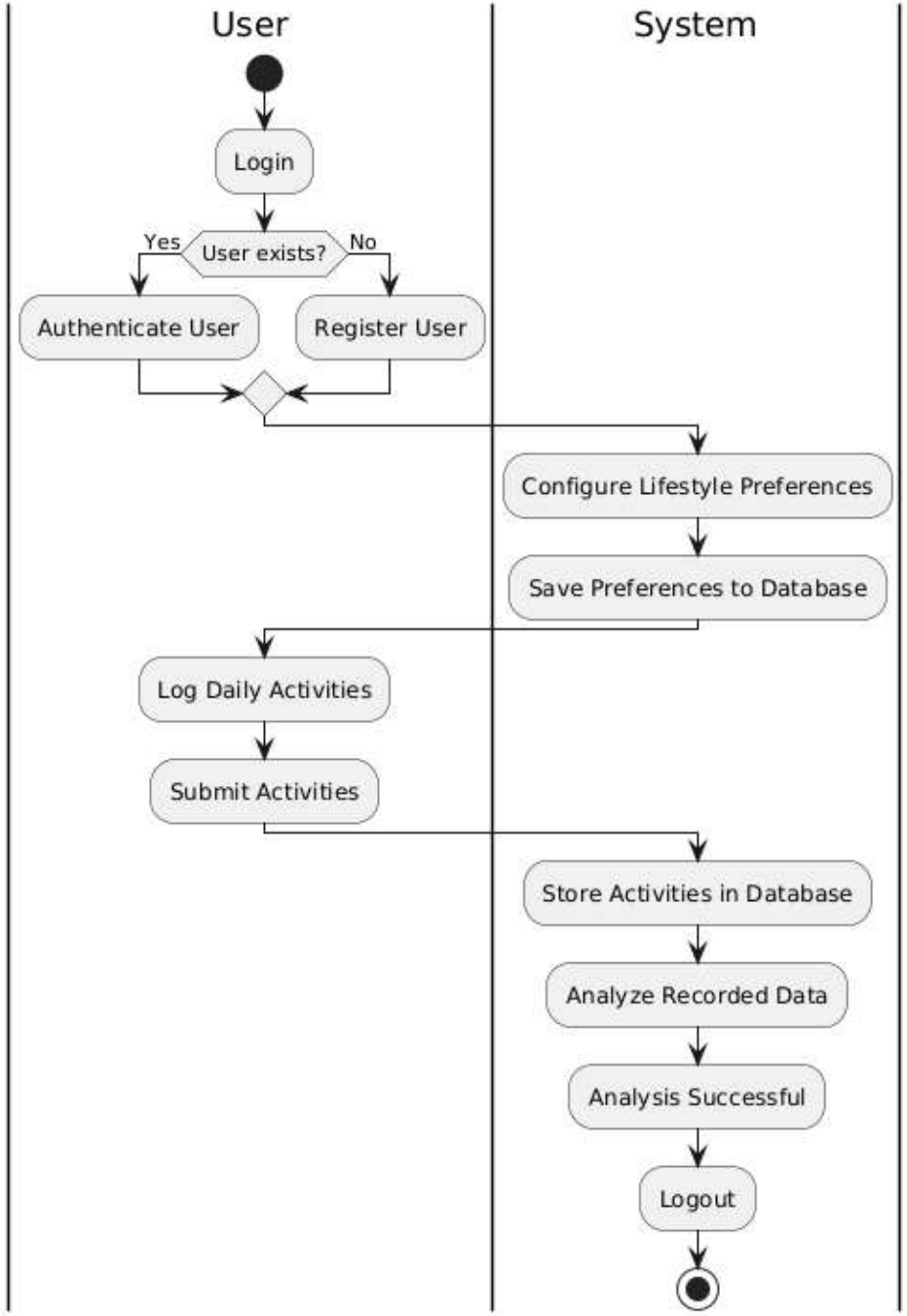
# Diagram

## Use Case Diagram



# Diagrams

## Activity Diagram




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# DEMONSTRATION

# Testing And Result

| | |
|---|---|
| White Box Testing<br><br>- Scope: Individual class functions and constructors.<br><br>- Responsibility: Developers test and debug their own Code.<br><br>- Approach: Verify each method's functionality. | Black Box Testing<br><br>- Scope: Integrated system after all components are assembled.<br><br>- Focus: Validate Tracker application under all possible usage scenarios.<br><br>- Comprehensive coverage of user interactions and workflows. |

# Future Enhancements

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- Educational Resources
- Gamification Elements
- Custom Reports
- Peer Comparison
- API for Data Import/Export -
  - Enable integration with fitness apps, GPS devices etc.

# Challenges

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- Synchronization issues with the database.
- Integrating APIs for data like local electricity grids, transport footprints.

# Highlights

- Monitor, analyse, and optimize daily activities for a sustainable lifestyle
- Track CO2 footprints in mobility, eating habits, and consumer behaviour
- Gain insights into carbon footprint
- Awareness: Understand daily CO2 footprints
- Personalized Insights: Tailored recommendations
- Data Visualization: Clear charts and reports
- Continuous Improvement: Make iterative improvements.

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# VIELEN DANK!

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